

Problem 27

A stone is dropped into a chasm and falls freely. The sound of the stone hitting the water is heard after a time of $t = 4.9$ s from the moment it was released. What is the depth of the chasm? Assume the speed of sound is $c = 340$ m/s.

Solution

Let the depth of the chasm be H . The total time t consists of two parts:

- t_f : the time for the stone to reach the water.
- t_s : the time for the sound to travel back up.

Thus,

$$t = t_f + t_s.$$

From the free-fall equation:

$$H = \frac{1}{2}gt_f^2.$$

The time for the sound to travel back is:

$$t_s = \frac{H}{c}.$$

Substituting into the total time equation*:

$$4.9 = \sqrt{\frac{2H}{9.81}} + \frac{H}{340}.$$

Define $x = \sqrt{\frac{2H}{9.81}}$, then $H = \frac{9.81x^2}{2}$. Rewriting the equation*:

$$4.9 = x + \frac{9.81x^2}{2 \times 340}.$$

Solving numerically,

$$H \approx 316 \text{ m}.$$

Thus, the depth of the chasm is approximately 316 m.